Jimmy Nguyen

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Education

Georgia Institute of Technology *Expected Jun 2028*
Master of Science in Computer Science | Specialization: Artificial Intelligence
Coursework:*Game AI, Introduction to Graduate Algorithms, Software Development Process*

University of California Davis *Jun 2024*
Bachelor of Science in Computer Science GPA: 3.7
Coursework:*Data Structures and Algorithms, Algorithm Design and Analysis, Operating Systems, Computer Architecture, Programming Languages, Computer Vision, Machine Learning, Artificial Intelligence*

Work Experience

Apple Career Experience *Jun 2026 - Nov 2026*
Software Support Engineer Sunnyvale, CA

- **Designed and built a full-stack internal web application** (Python, **FastAPI**, PostgreSQL) used across retail locations to manage and provision hardware devices, replacing a manual process with **self-service**.
- **Implemented secure single sign-on** (OAuth/OIDC) with **role-based access control**, and deployed the app to a **Kubernetes**-based platform via automated **CI/CD pipelines** and GitOps-style delivery.
- **Diagnosed and resolved production reliability issues**, including a memory-exhaustion crash and a gateway-timeout failure, by introducing **concurrency controls** and infrastructure configuration fixes.
- **Built a Python-based QA automation framework** combining UI automation with **AI-driven visual validation** to test iOS/iPadOS applications across 10+ locales, **cutting manual regression testing time**.
- **Automated bug-triage workflows** by building a scheduled service that parsed structured internal bug reports and orchestrated backend actions automatically, and filed detailed, reproducible defect reports.

Projects

Pokémon Sealed Inventory Tracker | React | Python | FastAPI | SQLite *Jan 2026 – Nov 2026*

- Built a **full-stack portfolio tracker** with **gain/loss analytics**, allocation breakdowns, and a **24-month value history** fed by an external pricing API.
- **Cut API latency by up to 74%** (74 → 19 ms) at 10K holdings and the holdings endpoint by **64%** by moving aggregation into **SQL** and grouping holdings before valuation.
- Developed a **FastAPI** REST backend on **SQLite** with a price-history model, deploying the frontend to **GitHub Pages** and the API to **Render**.
- Reached **99% backend / 90% frontend test coverage** (46 tests), enforced in **GitHub Actions** CI.
- Designed a **React** dashboard with custom SVG charts, scoring **Lighthouse 100** in all categories.

Credit Score Explorer | Python | scikit-learn | Streamlit | pandas *Mar 2025 – Nov 2026*

- **Led a team of 5** to build a credit score classifier on 100K records, later **rebuilt** as a deployed, **explainable ML web app** with a reproducible training pipeline.
- Designed an **ordinal gradient-boosting model with monotonic constraints**, guaranteeing that more debt or late payments can never improve a predicted score.
- Built **model-agnostic explanations** and a **what-if search** that finds the smallest change to raise a user's score band (e.g., debt $3,400 → $1,200).
- Achieved **66% accuracy** vs. a 53% baseline on **held-out customers**, using a customer-level split to prevent the data leakage that inflates row-level splits.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, HTML/CSS, Java, Erlang, Matlab

Tools & Technologies: React, FastAPI, scikit-learn, TensorFlow, pandas, Streamlit, SQLite, PostgreSQL, Git, Docker, Kubernetes, REST APIs, CI/CD (GitHub Actions), OAuth/OIDC, Playwright, Agile/Scrum workflows